
Running the Theory: An Empirical Study of Model-Based Reinforcement Learning for Confounded POMDPs

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Abstract

Offline reinforcement learning is systematically biased when the behavior policy acted on latent state the learner never observes. Hong et al. (2024) give a model-based theory for this setting, identifying reward and transition models through two families of *bridge functions* and planning pessimistically over a confidence region. The paper is theory only: no algorithm, no code, no experiments. We build the pipeline end-to-end, verify it against exact ground truth to machine precision, and report what running the theory reveals. Three findings stand out. First, the paper’s pessimism step is numerically unusable as written: its joint multi-block minimization diverges to -1779 against true values in $[1.5, 2.2]$, and we trace the mechanism to a structural null space in the design matrix. Second, a signal-subspace projection repairs the numbers and achieves zero selection regret, but we show it forfeits coverage — the true bridge leaks 21% of its norm outside the projected subspace — making it a disclosed heuristic rather than a certified bound. Third, at benchmark scale a confounding-blind baseline beats every bridge-based method in all 15 sweep cells, worst at *zero* confounding, identifying estimation variance rather than confounding bias as the binding constraint. Extending to continuous spaces with kernel bridges preserves validity but not selection quality, and a diagnostic in the paper-faithful configuration (continuous states, *finite* actions) explains why: the pessimism penalty is monotonically inverse to behavior-policy coverage, so pessimism systematically avoids the optimal policy exactly when the logging policy under-covers it.

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1. Introduction

Offline RL from logged data is unreliable when the logging policy observed more than the learner does. In a partially observable MDP whose *latent* state drives both the recorded actions and the outcomes, the data are *confounded*: treating the observation as the state yields a bias that does not vanish with more data. This is the central obstacle to trustworthy offline RL in healthcare and operations, where unrecorded drivers of decisions are the norm rather than the exception.

Hong et al. (2024) address exactly this setting. Using a pre-decision *negative control* O_0 , they identify the confounded POMDP’s reward and transition models through two families of bridge functions (b_R, b_D), then plan pessimistically over a policy-independent confidence region. The result is elegant and, importantly, stated for general spaces: states and observations may be continuous, and general function approximation is not merely permitted but required. The paper is, however, *theory only* — it contains no algorithm, no code, and no experiments. Its neighbors are likewise unrealized empirically for this exact pipeline: the closest follow-up runs the bridge *estimator* in isolation (Venkatesh & Malikopoulos, 2026), and the pessimism precedent is theory-only and for the *unconfounded* case (Guo et al., 2022).

This leaves a concrete question that only implementation can answer: *when the framework is built end-to-end, does model-based bridge identification actually de-bias policy evaluation at practical sample sizes, and can its abstract pessimistic max–min step be made to run at all?*

Contributions.¹

1. An end-to-end implementation estimating *both* bridge families and carrying them through to a tractable pessimistic policy-*selection* step, verified against exact oracles to $\sim 10^{-15}$ (Section 4.1) — a capability claim we can check ourselves, not a primacy claim.
2. A *diagnosis* of a failure mode in the paper’s coupled

¹Code and raw results: github.com/soheilsayahvarg/model-based-rl. The benchmark builds on the public *gumbel-max-scm* simulator (Oberst & Sontag, 2019).

multi-block pessimism, plus a signal-subspace repair and a quantified account of the coverage it forfeits (Section 3.3, Section 4.3).

3. A characterization of *where the approach stops working*: a naive baseline wins all 15 cells of a benchmark sweep, worst where there is nothing to correct (Section 4.4).
4. A kernel extension to continuous spaces, and — in the paper-faithful configuration with finite actions — a diagnostic identifying *behavior-policy coverage* as the mechanism behind pessimistic selection failure (Section 4.5, Section 4.6).

2. Related Work

Proximal causal inference. Bridge functions originate in the double negative control literature (Miao et al., 2018; Tchetgen Tchetgen et al., 2020), where a pair of proxy variables identifies a causal effect under unmeasured confounding. Mastouri et al. (2021) give the kernel two-stage estimation procedure our continuous extension instantiates.

Model-free OPE in confounded POMDPs. Shi et al. (2022) and Bennett & Kallus (2024) identify policy values through *value* bridge functions solved from Bellman-type backward moment equations. These depend on the evaluation policy, so a new estimation problem is solved per policy; the model-based route of Hong et al. (2024) instead extracts policy-independent reward and transition information, which is what makes a policy-independent confidence region possible. We implement Shi et al. (2022) as a competing baseline.

Pessimism in offline RL. Conservative planning over confidence regions is standard in fully observed offline RL and has a POMDP precedent in Guo et al. (2022), which is theory-only and unconfounded. The failure mode of Section 3.3 is specific to the interaction between a structural null space and *joint* multi-block minimization.

3. Method

3.1. Identification

We adopt the setting of Hong et al. (2024): a finite-horizon POMDP with latent states S_t , observations O_t , actions A_t and rewards R_t , where the behavior policy may depend on S_t . A pre-decision negative control O_0 satisfies $O_0 \perp (O_t, O_{t+1}, R_t) \mid S_t, A_t, H_{t-1}$. Reward-emission and dynamic-emission bridge functions solve the conditional moment restrictions

$$\mathbb{E}\left[b_R^{[t]}(A_t, O_t, r_t, o_t) \mid H_{t-1}, A_t, O_0\right] = p(r_t, o_t \mid H_{t-1}, A_t, O_0) \quad (1)$$

and analogously for $b_D^{[t]}$ with target $p(o_{t+1}, o_t \mid \cdot)$. Under a completeness condition, the policy value of *any* history-dependent policy is identified by a sequential contraction of these bridges (Hong et al., 2024, Thm. 3.5). We implement this contraction exactly and verify it against dynamic programming on the latent model.

3.2. Estimation

Stage 1 fits a conditional mean embedding $\hat{\mu}_{W_t|x}$ by kernel ridge regression; Stage 2 solves the resulting empirical operator equation in closed form, $\alpha = (G + N_2 \lambda_2 I)^{-1} \hat{p}$. We implement both the dual (representer) and primal (operator) forms and verify they agree to 2.4×10^{-15} , which catches derivation errors that no single implementation could.

Regularization does not transfer. The paper’s RKHS rate schedules, applied literally to a tabular instantiation, over-shrink the estimator by $\approx 80\%$ by the third time-step. We grid-calibrate instead: a 0.7:0.3 split with $\lambda_1 = 1/N_1$ and $\lambda_2 = 0.03/\sqrt{N_2}$. These values are measured, not derived.

The usual ill-posedness diagnostic misleads. Because the observation space is structurally richer than the latent space ($|O| > |S|$), the design matrix carries a null space that pins $\text{cond}(G)$ flat even as estimation quality degrades tenfold — confirmed by an O_0 -noise sweep in which Stage-1 error grows $10\times$ while $\text{cond}(G)$ does not move. We monitor the smallest *signal* eigenvalue instead.

3.3. Pessimism, its failure mode, and a repair

Confidence regions are exact ellipsoids centered at $\hat{b}$, and the per-block inner minimum is the standard ellipsoid support-function identity, $V(\hat{b}) - \sqrt{\xi} \|g_\pi\|_{H^{-1}}$. This is not new mathematics and we do not claim it as a contribution. The difficulty is the paper’s *coupled* minimization across blocks.

Failure mode. Run as written, the joint minimization exploits precisely the null-space directions identified above: they cost the confidence-region geometry almost nothing, so the optimizer walks arbitrarily far into them and the pessimistic value diverges (Section 4.3). The diagnosis and the spectral-monitor fix share one root cause.

Signal-Projected Pessimism. We restrict the pessimistic search to the empirically identified signal subspace via an SVD/eigengap decomposition with orthonormal basis U and $H_{\text{sub}} = U^\top H U$. This restores informative values and correct selection — but, as Section 4.3 shows, at a coverage cost we quantify rather than assume away.

Table 1. Oracle verification. Every downstream result is therefore a fact about the method, not an artifact of the code.

Check	Max gap
Policy-value chain vs. exact DP	4.4×10^{-16}
Estimator vs. oracle bridges (population)	3.0×10^{-15}
Dual vs. primal solve (independent derivations)	2.4×10^{-15}
Analytic vs. finite-difference gradients	4.9×10^{-10}

3.4. Continuous extension

The framework is not inherently tabular: Hong et al. (2024) assume continuous S and O and call function approximation *inevitable* there. We instantiate the same pipeline with a two-stage **kernel** estimator on a linear-Gaussian confounded POMDP admitting an exact closed-form oracle, reusing the pessimism engine unmodified since it works purely in coefficient space (derivation: Appendix B).

4. Experiments

All multi-seed results use 5 seeds with 95% confidence intervals. Environments: a minimal tabular POMDP ($|S|=2, |A|=2, |O|=3, T=3$) with closed-form oracle bridges; a 720-observation benchmark built on the public `gumbel-max-scm` clinical simulator (Oberst & Sontag, 2019) with a manufactured O_0 satisfying the identification assumption by construction; and the continuous linear-Gaussian POMDP of Section 3.4. Everything runs on a laptop in NumPy; no GPU.

4.1. Correctness

We first pin the implementation against exact ground truth (Table 1). The dual-versus-primal check is the most informative: the two solves are independent derivations of the same estimator, so agreement to 2.4×10^{-15} rules out derivation errors a single implementation would silently carry. This is what licenses reporting the negative results below as findings rather than bugs.

4.2. De-biasing works where identification is clean

On the toy environment the plug-in evaluator beats the confounding-blind baseline in **12/12** (N, π) cells. Worst-policy plug-in bias shrinks $0.104 \rightarrow 0.053 \rightarrow 0.036$ as N grows from 5k to 50k, while naive bias stays flat at 0.13–0.33 regardless of N : the textbook signature of removed versus irreducible bias.

4.3. The pessimism step, run literally, diverges

A “pessimistic lower bound” of -1.15 is already outside the physically feasible range, and it degrades from there; selection regret is 0.68. This is not a tuning artifact — it

Table 2. Vanilla pessimism (the paper’s Eq. 17) at increasing region size c . True policy values lie in $[1.5, 2.2]$.

c	0.1	0.3	1.0	3.0	10.0
Best-policy V_{low}	-1.15	-10.4	-62.2	-305.2	-1779

Table 3. Benchmark sweep (5 confounding strengths $\times$ 3 sample sizes $\times$ 5 seeds). MAE against exact ground truth.

Method	MAE range	Beats naive
Naive (confounding-blind)	0.013–0.074	—
Model-based (plain)	0.072–0.230	0/15
Model-based (shrunk)	0.063–0.240	0/15
Model-free proximal	0.164–0.180	0/15

reproduces across seeds and we traced the mechanism to the null-space geometry of Section 3.3.

Signal-Projected Pessimism repairs it: $V_{\text{low}} = 1.35$ at $c = 0.03$, in range, with **0.000 selection regret** across every seed.

The honest limit. We then stress-tested our own fix. The true bridge functions leak up to **21% of their norm** outside the projected signal subspace, so the projected region provably does *not* contain the truth — its honest coverage is zero. We therefore describe the situation as a **coverage/informativeness trade-off**: vanilla pessimism covers the truth but is uninformative; our projection is informative and empirically correct but uncertified. $V_{\text{low}} \leq V_{\text{true}}$ held in every test we ran, by empirical sign-alignment rather than by guarantee.

4.4. At scale, a naive baseline wins

The de-biasing method’s *worst* performance is at $\kappa = 0$, where it is 2.8–6.8 $\times$ worse than naive — precisely where there is nothing to correct. That is the clearest possible signature that the dominant error source at this scale is **estimation variance**, not the confounding bias the method targets: inverting many small, noisy per-cell matrices injects more noise than the bias it removes. The model-free comparator is flat in N , which we confirmed is an identification limit of the benchmark’s binary proxy rather than a bug — it *is* consistent on the toy (0.0076 max error at $N=200k$).

4.5. Continuous state, action and observation spaces

Bridge recovery improves with sample size ($0.220 \rightarrow 0.055$ reward, $0.240 \rightarrow 0.102$ dynamics, for $N : 300 \rightarrow 4,000$), and plug-in de-biasing beats naive in **4/4** policies by 3–11 $\times$ — the continuous analogue of the tabular 12/12, with the same signature in the intervals: naive bias tight but systematically wrong, plug-in bias smaller but noisier.

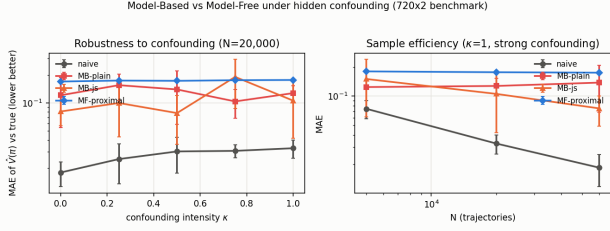


Figure 1. Naive vs. model-based (plain/shrunk) vs. model-free, 5-seed 95% CIs. Left: vs. confounding strength κ at $N=20,000$. Right: vs. sample size at $\kappa=1$. Naive stays lowest throughout.

Table 4. Continuous kernel extension, 5 seeds, mean $\pm$ 95% CI at $N=2,000$. Oracle self-consistency 1.665×10^{-16} ; gradient check $\approx 10^{-11}$.

Policy	Naive bias	Plug-in bias
never_treat	-1.255 ± 0.018	-0.401 ± 0.216
constant_dose	$+0.093 \pm 0.013$	$+0.008 \pm 0.032$
proportional	-0.435 ± 0.013	-0.129 ± 0.061
aggressive	$+1.357 \pm 0.038$	$+0.365 \pm 0.178$

Pessimism validity holds throughout, but **validity is not selection quality**: ranking by V_{low} is consistently worse than ranking by the plug-in value (regret 0.40–0.48 versus 0.241), unlike the tabular case’s clean 0.000. Two bottlenecks are regime-specific: dense kernel ridge costs $O(N_1^3)$, capping $N \approx 4,000$; and a *finite-dimensional* kernel silently re-creates the rank deficiency of Section 3.3, caught only by checking the Gram matrix’s rank, since recovery numbers looked correct either way (Appendix B).

4.6. The paper-faithful configuration, and why pessimistic selection fails

Hong et al. (2024) assume continuous S and O but an explicitly *finite* action space, so Section 4.5 varies two things at once. We repeated the study with $A = \{0, 0.5, 1\}$, giving each action level its own bridge coefficient (a delta kernel on the action coordinate tensored with a linear kernel on the observation), all else fixed. The oracle check is exact (0.000), recovery is monotone in N , and validity again holds at 100%.

The anomaly persists, refuting our own hypothesis that continuous actions caused it. It is also not an optimization failure: restart gaps here are $\leq 4 \times 10^{-16}$ up to $c=1$, so coordinate descent is finding the global optimum. Table 5 identifies the actual mechanism. The pessimism penalty is monotonically inverse to how well the behavior policy covers each candidate, and in this environment the confounded logging policy prefers high doses (62% of actions) while the optimal policy never treats (12%). Pessimism is therefore not malfunctioning — it is doing its job, penalizing policies the data does not support, and here the optimal policy is the unsupported one. The account predicts the gap should close

Table 5. Pessimism penalty $V_{\text{plugin}} - V_{\text{low}}$ against behavior-policy coverage, finite-action setting ($N=2,000$, $c=0.3$). The ordering is monotone, and the *best* policy is the *least* covered.

Policy	V_{true}	Penalty	Coverage
never_treat (best)	1.544	2.058	0.121
low_dose	1.243	1.350	0.260
mixed	1.333	1.092	0.262
high_dose (worst)	0.942	0.842	0.619

as coverage of the optimal action improves.

De-biasing here wins 2/4 rather than 4/4, consistently: where naive bias is already negligible (0.008, 0.009 for the well-covered policies) there is nothing to correct and the correction only adds variance — Section 4.4 in miniature.

5. Limitations

The pessimism guarantee is empirical. The signal projection provably excludes part of the truth (21% norm leakage), so $V_{\text{low}} \leq V_{\text{true}}$ is an observation across our tests, not a theorem.

The coverage explanation rests on one environment. Table 5 gives a monotone four-point ordering plus a mechanism, which we find convincing, but a proper test would sweep behavior-policy coverage of the optimal action directly and check that the selection gap closes as predicted. We have not run that sweep.

Restricted scope. Continuous-action candidates are affine and observation-only; finite-action candidates are observation-independent, which is what preserves the exact oracle. All experiments use $T=3$, scalar continuous spaces, and $N \leq 4,000$. Our primary comparator is confounding-blind; the one confounding-aware comparator is under-identified by the benchmark’s binary proxy, so a second would sharpen Section 4.4.

Statistical self-correction. Two earlier headline numbers rested on 3-seed standard deviations; re-checking at 5 seeds changed one materially and refuted the other outright. Both are recorded, with the retracted mechanism for Section 4.6, in Appendix C.

6. Conclusion

Running a theory-only framework end-to-end showed four things the theory did not: its pessimism step does not run as written; the repair costs a measurable guarantee; a naive baseline dominates at scale through estimation variance, not confounding bias; and pessimistic selection fails because the penalty tracks behavior-policy coverage rather than value.

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A. Experimental Setup

Environments. *Toy POMDP:* $|S|=2$, $|A|=2$, $|O|=3$, $T=3$, with closed-form oracle bridge functions obtained by pseudo-inverse and exact dynamic-programming policy values. This is the only substrate where recovery error against ground truth is directly measurable, and it carries every machine-precision correctness check.

Benchmark: 720 observable-core states $\times$ 2 latent states, 8 actions, $T=3$, built on the public `gumbel-max-scm` clinical simulator (Oberst & Sontag, 2019). The latent index drives dynamics and behavior but is censored from the logs. We construct a pre-decision negative control O_0 satisfying the identification assumption by construction, plus a per-step noisy marker that guarantees bridge existence — deterministic masking alone gives rank $720 < 1440$, i.e. no valid one-step bridge. A confounding knob $\kappa \in \{0, 0.25, 0.5, 0.75, 1.0\}$ sweeps from no confounding to strong confounding.

Continuous: scalar linear-Gaussian confounded POMDP, described in Appendix B.

Protocol. All multi-seed results use seeds $\{0, 1, 2, 3, 4\}$ and report $1.96 \hat{\sigma} / \sqrt{n}$ intervals. Ground truth throughout is exact dynamic programming on the latent SCM (tabular) or a closed-form recursion (continuous). The full pipeline runs on a laptop in NumPy; no GPU is used or required.

Scaling the benchmark estimator. A direct implementation of the paper’s history-conditioning would require on the order of billions of table cells at benchmark scale. Three exact structural factorizations specific to the environment’s observation design — hashed encoding, core-diagonal bridge support, and closed-form 2×2 local inversions — keep the pipeline laptop-scale without approximation.

B. Continuous Extension: Derivation and Details

B.1. Environment

State, action, observation and reward are all real-valued. The observation is a noisy proxy $o_t = s_t + \varepsilon_t$ for the hidden state, and the logging policy may act on the true s_t (confounded) or only on o_t (not confounded), mixed by a knob κ . This is the classical *measurement-error* setting that proximal causal inference was designed for: fitting a model that treats the noisy proxy as if it were the state induces errors-in-variables bias whenever $\kappa > 0$.

Because the reward is linear and candidate policies are affine and observation-only ($a = Ko + c$), the closed-loop mean trajectory is an exact scalar recursion — no covariance propagation is needed. This yields an exact closed-form oracle for both policy values and bridge coefficients, verified to reproduce each other to 1.665×10^{-16} .

B.2. Mean-value bridges

The paper’s bridge functions are *distributional*: they must match $p(y | X_t)$ for every y . Reproducing that in continuous y requires conditional density estimation. Since the downstream quantity is a policy-value *mean* and the pessimism machinery concerns estimation uncertainty in the bridge coefficients rather than the environment’s intrinsic reward noise, a conditional-mean moment restriction suffices and is exact for that purpose:

$$\mathbb{E}[b(A_t, O_t) | X_t = x] = \mathbb{E}[Y_t | X_t = x] \quad \forall x, \quad (2)$$

with $Y_t = R_t$ (reward bridge) or O_{t+1} (dynamics bridge). This is exactly a nonparametric instrumental-variable regression, with $W = (A, O)$ confounded, $X = (A, H_{t-1}, O_0)$ the instrument, and Y the outcome — well preceded in the proximal literature (Mastouri et al., 2021). It is a deliberate scope reduction, not a fundamental barrier.

B.3. Two-stage kernel solve

We derived the closed form directly from a stated objective rather than pattern-matching a remembered formula, because a mis-remembered index convention would silently produce a wrong-but-plausible estimator — the exact failure mode oracle verification exists to catch. With K_X , K_W the instrument- and bridge-side kernels,

$$\Gamma = (K_X(X_1, X_1) + N_1 \lambda_1 I)^{-1} K_X(X_1, X_2), \quad (3)$$

$$Z = K_W(W_1, W_1) \Gamma, \quad (4)$$

$$H = \frac{1}{N_2} Z Z^\top + \lambda_2 K_W(W_1, W_1), \quad g = \frac{1}{N_2} Z y_2, \quad (5)$$

and $\beta = H^{-1}g$. **Unifying check:** when K_W is a one-hot/delta kernel (the tabular case), $K_W(W_1, W_1) = I$ exactly and this reduces to the tabular estimator’s own primal solve — a correctness check on the derivation itself, not merely an analogy.

B.4. A rank-deficiency pitfall

For the primary verified path we take K_W to be a *linear* kernel, since the true bridges are exactly linear and a linear kernel’s RKHS therefore contains them. But a linear kernel has a finite (2-dimensional) feature space, so the $N_1 \times N_1$ Gram matrix $K_W(W_1, W_1)$ has rank ≤ 2 — confirmed empirically, with the remaining eigenvalues at the $\sim 10^{-13}$ floor. The dual system is therefore numerically meaningless in those directions.

Notably, the specific projected quantity we originally extracted ($\hat{\theta} = W_1^\top \beta$) turned out to be *invariant* to that instability and produced identical recovery numbers before and after the fix — so the bug was invisible from the recovery metrics. It still had to be fixed, because H and $\hat{b}$ enter the pessimism ellipsoid directly, where no such invariance protects them. The fix is to solve in the 2-dimensional *primal* coordinates: mathematically identical, exactly well-conditioned, and incidentally $\sim 1000\times$ cheaper.

B.5. Spectral monitoring

The tabular fix for a misleading condition number was a hard eigengap on the smallest signal eigenvalue, correct there because the null space is *structural* ($|O| > |S|$, exactly zero signal beyond it). A continuous kernel Gram matrix has smooth spectral decay rather than a structural wall, so a hard eigengap does not transfer. We use the standard kernel-ridge *effective dimension* $N_{\text{eff}}(\lambda) = \text{tr } K(K + N\lambda I)^{-1}$ (Caponnetto & De Vito, 2007) instead.

B.6. Gradients

The value chain is multilinear in each stage’s coefficients, so reward-block gradients are exact. A dynamics block’s coefficients additionally shift the *evaluation point* of every later stage, and that dependence runs through the kernel nonlinearly; we keep the coefficient-linear part exact and obtain the one remaining scalar sensitivity by a single 1-D central difference — $O(T)$ extra work per gradient call, not $O(N_1)$ — verified against numerical differentiation of the whole chain to $\approx 10^{-11}$.

B.7. Regularization: a tested-and-rejected hypothesis

We hypothesized that the tabular-calibrated $\lambda_1 = 1/N_1$ would be too weak for a genuine kernel conditional mean embedding, since the measured effective dimension came out close to N_1 itself. We tested $1/\sqrt{N_1}$ and $5/\sqrt{N_1}$ with 5-seed averaging: both performed *worse*. We kept $1/N_1$. This is the mirror image of the tabular finding — there, the paper’s RKHS schedule failed to transfer to a tabular setting; here, the tabular-calibrated schedule transferred fine to a genuine kernel.

B.8. Full pessimism sweep

c	0.03	0.1	0.3	1.0	3.0
Fraction valid ($V_{\text{low}} \leq V_{\text{true}}$)	1.00	1.00	1.00	1.00	1.00
Pessimistic regret	0.479	0.438	0.397	0.397	0.397
Plug-in regret	0.241	0.241	0.241	0.241	0.241
Max restart gap	2.2e-16	1.36	5.26	20.4	62.2

Validity holds everywhere. Selection quality does not match the tabular case, and the restart gap grows sharply with region width — consistent with the optimization landscape becoming genuinely harder to solve exactly, rather than merely more conservative.

C. Corrections to Our Own Results

We record these because the verification process producing them is part of the contribution, not an embarrassment to be smoothed over.

Correction 1: harm ratio at zero confounding. On a 3-seed sweep we reported the model-based estimator as $23\text{--}46\times$ worse than naive at $\kappa = 0$. At 5 seeds the ratio is $2.8\text{--}6.8\times$. Still a real, clear negative result — just a substantially smaller one than we had stated.

Correction 2: retracted variance-reduction claim. We had claimed James–Stein shrinkage “cuts run-to-run variance sixfold under strong confounding.” It does not replicate. Across the full 15-cell sweep, shrinkage has lower variance than the plain estimator in only 7/15 cells — roughly a coin flip — and at $\kappa = 1$ specifically, the exact condition cited, its variance is *higher* at $N=5,000$ and $N=20,000$. The claim is retracted. The honest summary is a modest, noisy $\approx 7\%$ average MAE reduction, not a variance-reduction guarantee.

Neither was a coding bug: both were correctly computed standard deviations over an $n=3$ sample, which is too few draws to be stable. The lesson we take forward is that a claim derived from a *standard deviation* needs more seeds than a claim about a mean alone.

Correction 3: scope of the continuous study. We initially described the continuous extension as testing “the paper’s own generality.” Re-reading Hong et al. (2024) showed it assumes continuous S and O but an explicitly *finite* action space, while our environment uses a real-valued action. The oracle check still passes exactly, but the work is extrapolation beyond the stated theorem rather than an instantiation of it. We corrected the framing rather than relabel the experiment, and then ran the paper-faithful configuration (Section 4.6).

Correction 4: retracted mechanism for the selection failure. We had hypothesised that pessimistic selection underperformed because worst-case perturbations compound through the $T=3$ chain, citing coordinate-descent restart gaps that grew with region width. The finite-action experiment refutes this on two counts: the anomaly persists when the action space is finite, and restart gaps there are $\leq 4 \times 10^{-16}$ up to $c=1$, so the optimizer is finding the global minimum. The mechanism is instead behavior-policy coverage (Table 5). We report the failed hypothesis rather than silently substituting the new explanation, because the sequence — hypothesis, targeted experiment, refutation, replacement — is what makes the final claim credible.

D. Finite-Action Experiment: Full Results

Environment: continuous scalar state and observation, $A = \{0, 0.5, 1.0\}$, $T=3$, behavior policy a softmax over dose levels driven by the true hidden state (so overlap holds by construction). Candidate policies are observation-independent distributions over A ; that restriction is what preserves the exact oracle, since the expected bridge feature under such a policy is exactly $[\pi, m_t]$ and the chain stays affine in the mean.

Two implementation improvements over Appendix B follow from the finite action space. The bridge feature map $[\text{onehot}(a), o]$ is $(K+1)$ -dimensional, so the primal solve is mandatory from the outset rather than discovered. And because the chain is affine in the mean, dynamics-block gradients become *exact* via the backward recursion $D_t = b_{R,t}[-1] + b_{D,t}[-1] D_{t+1}$, replacing the finite difference of Appendix B; verified against numerical differentiation to 1.16×10^{-10} .

N	300	800	2,000	4,000
Reward-bridge error	0.147	0.141	0.083	0.071
Dynamics-bridge error	0.271	0.251	0.156	0.086

Recovery is monotone in both families, cleaner than the continuous-action run, which was non-monotone at an intermediate point.

c	0.03	0.1	0.3	1.0	3.0
Fraction valid	1.00	1.00	1.00	1.00	1.00
Pessimistic regret	0.211	0.211	0.289	0.602	0.602
Plug-in regret	0.000	0.000	0.000	0.000	0.000
Max restart gap	0.0	1.1e-16	2.2e-16	4.4e-16	5.33

Plug-in selection is perfect at every width; pessimistic selection is not, and the near-zero restart gaps rule out an optimization explanation. See Table 5 for the coverage diagnostic that does explain it.

E. Bugs Found by Oracle Verification

Two implementation bugs in the model-free comparator were invisible from the derivation and surfaced only against exact ground truth. First, a per-stage direct-method implementation was silently biased off-policy: a policy with true value 1.865 evaluated to 1.61, with the error *not* shrinking in N — the fingerprint of a derivation error rather than noise. It was fixed with a proper backward recursion. Second, the corrected version naively scaled to the benchmark attempted to build and invert an $\approx 11,500 \times 11,500$ dense matrix per policy per fit (≈ 2 GB), which stalled the run entirely; exploiting a block-diagonal structure the naive implementation missed resolved it.

The rank-deficiency issue of Appendix B belongs to the same category, with the additional wrinkle that it was invisible even from the downstream recovery metrics.